**6** **(a)** The photoelectric effect provides evidence for the particulate nature of electromagnetic radiation. State two experimental observations that could not be fully explained using the classical wave theory.

1. .....

.....

..... .

2. .....

.....

..... [2]

**(b)** In an experiment to investigate the photo-electric effect, the wavelength of the radiation incident on the metal surface was varied. For two values of wavelength $\lambda$, the stopping potential $V_s$ was measured. The results are shown in Fig. 6.1.

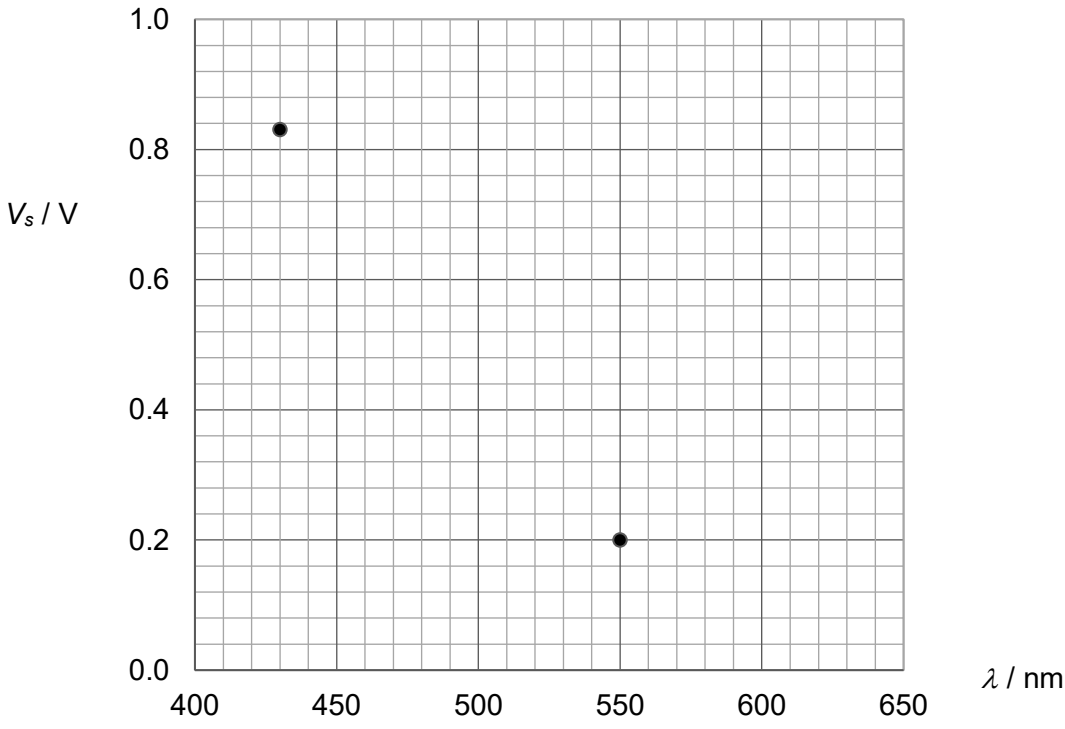


**Fig. 6.1**

**(i)** Determine the maximum kinetic energy of a photo-electron emitted from the metal surface by radiation of wavelength 550 nm.

maximum kinetic energy = .....J [1]

**(ii)** Hence, calculate the threshold wavelength of the metal.

threshold wavelength = ..... m [2]

**(iii)** Suggest why it is not possible to deduce the threshold wavelength of the metal surface directly from Fig. 6.1.

.....

..... [1]

**(iv)** The intensity of the radiation incident on the metal surface was kept constant as the wavelength was decreased from 550 nm to 430 nm.

State and explain the effect, if any, on the photocurrent.

.....

.....

..... [2]